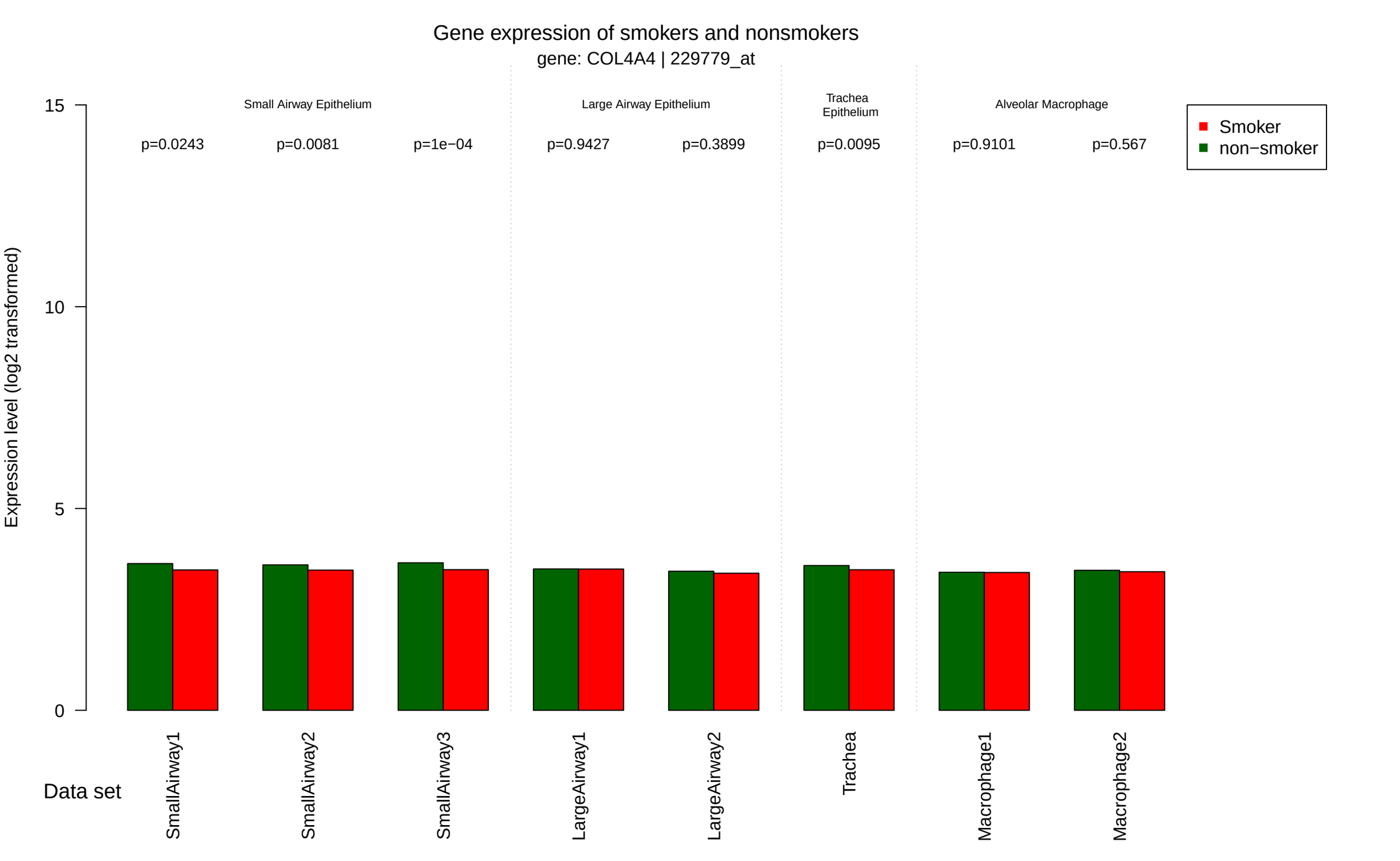
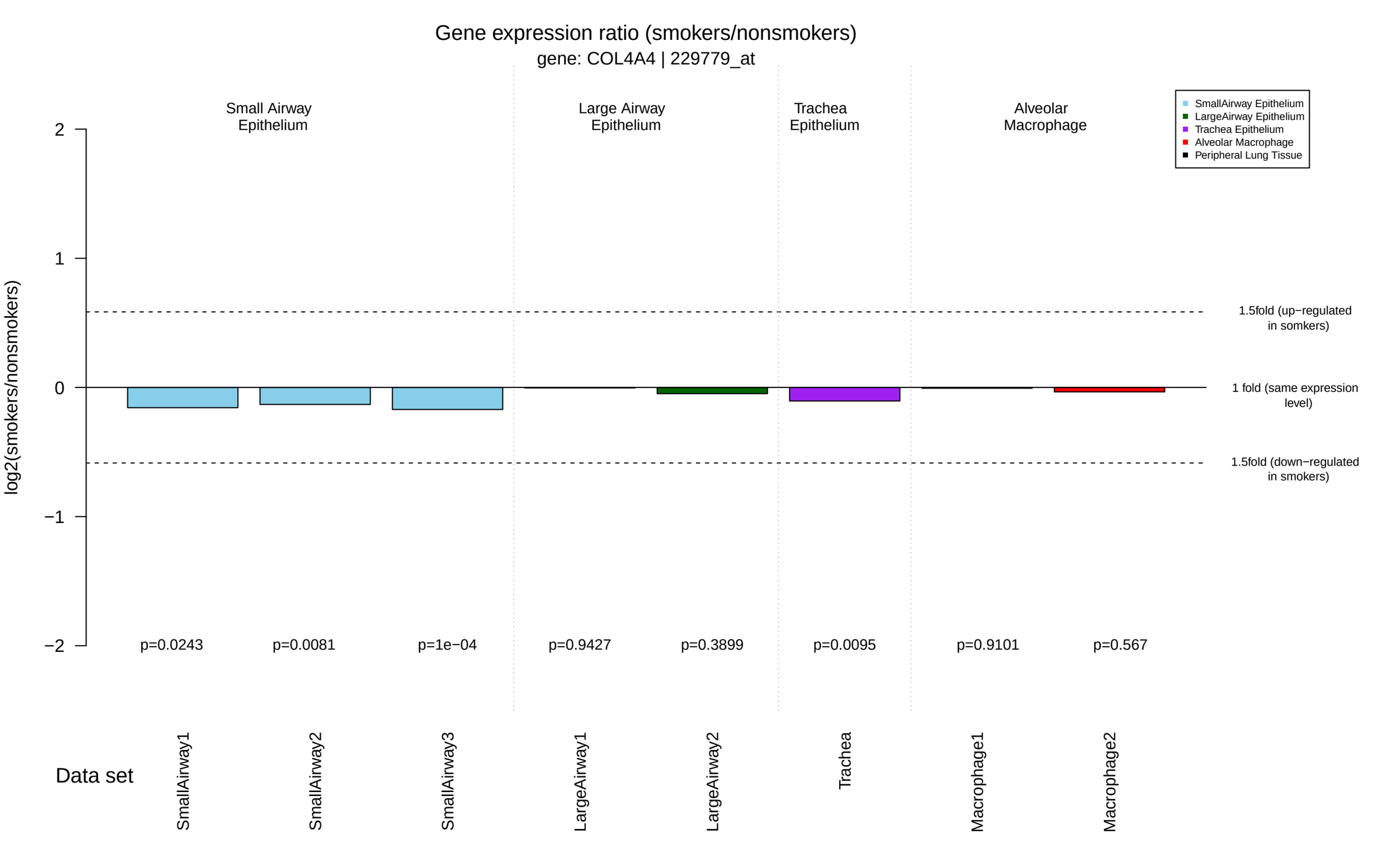
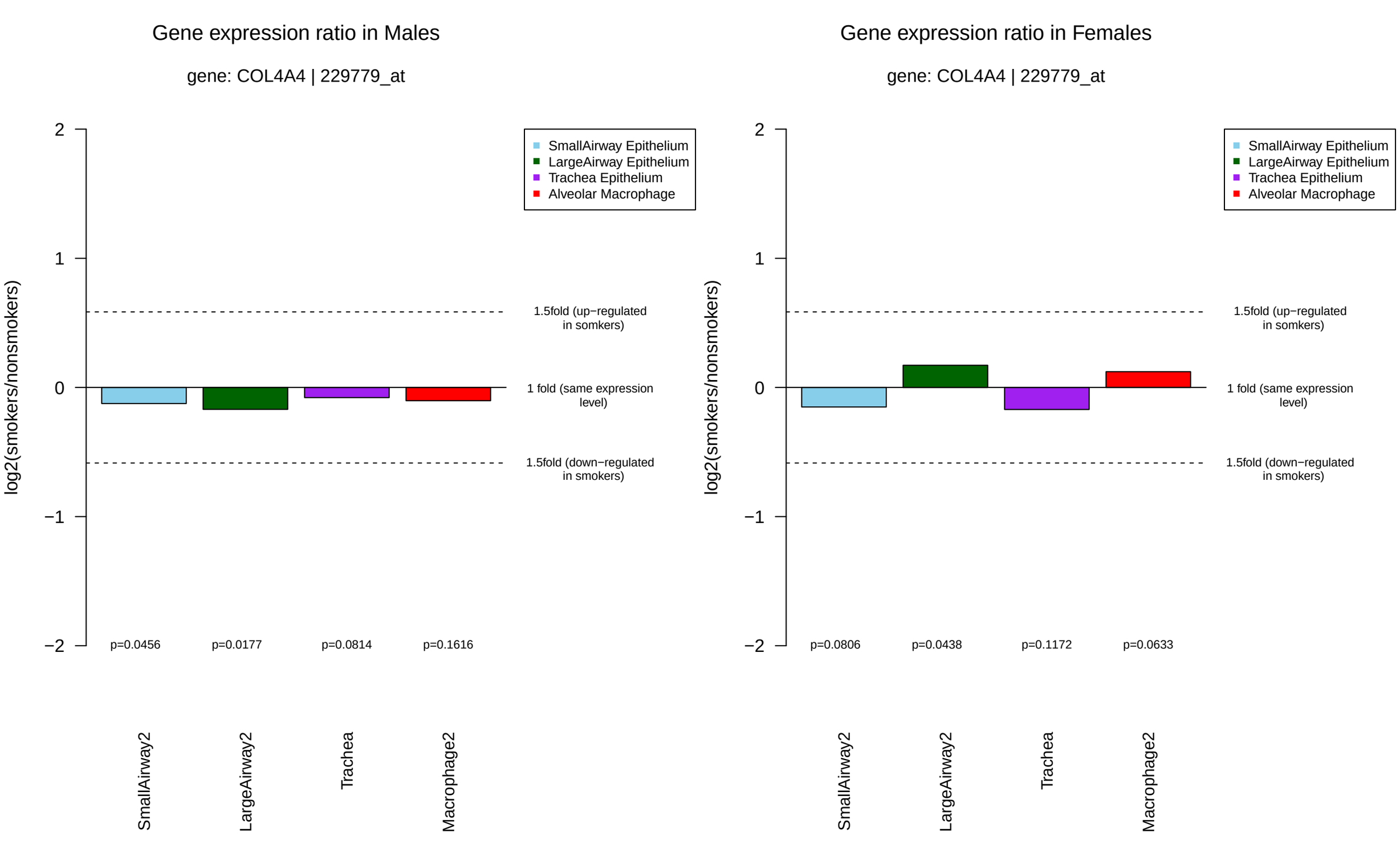
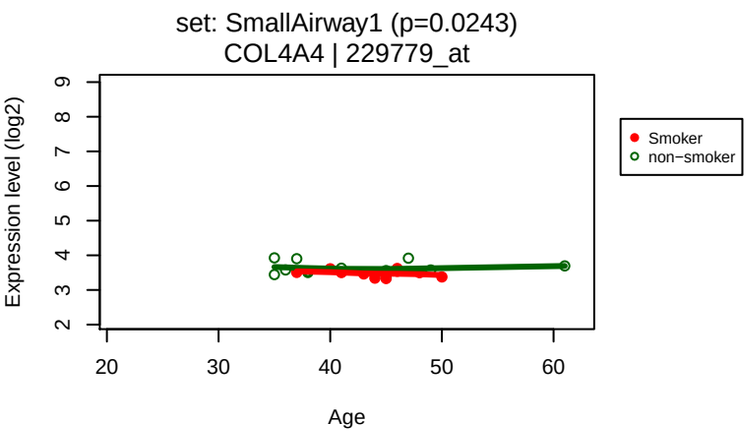
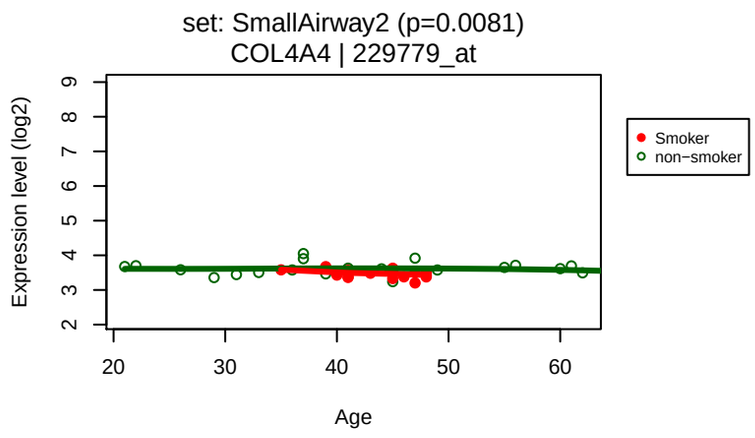
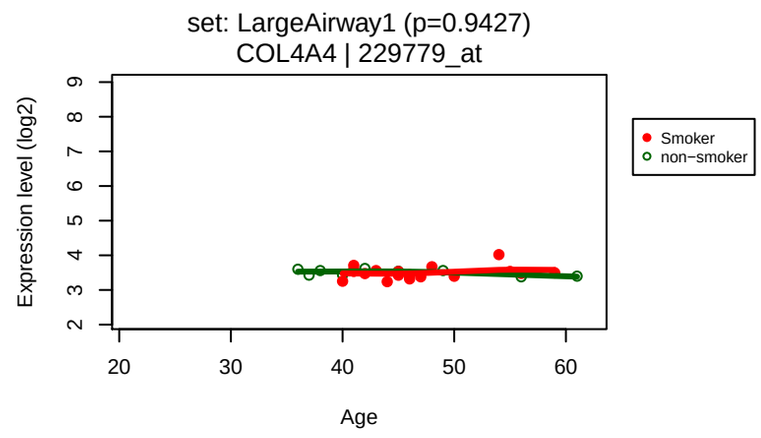

Gene expression of smokers and nonsmokers
gene: COL4A4 | 229779_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0243
p=0.0081
p=1e−04
p=0.9427
p=0.3899
p=0.0095
p=0.9101
p=0.567
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

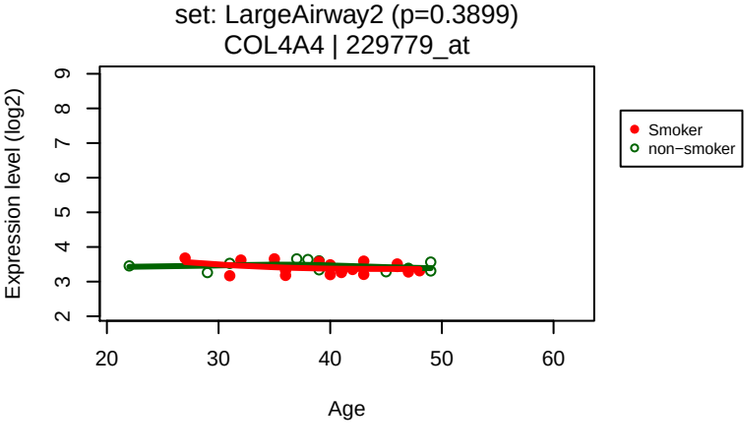

Gene expression ratio (smokers/nonsmokers)
gene: COL4A4 | 229779_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0243
p=0.0081
p=1e−04
p=0.9427
p=0.3899
p=0.0095
p=0.9101
p=0.567
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

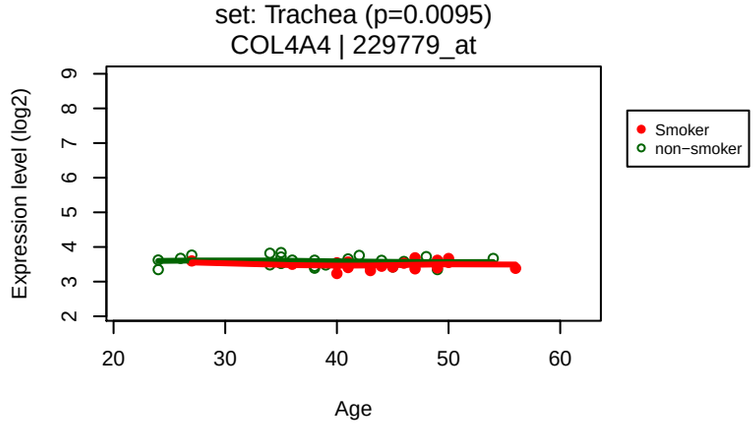

Gene expression ratio in Males
gene: COL4A4 | 229779_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0456
p=0.0177
p=0.0814
p=0.1616
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A4 | 229779_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0806
p=0.0438
p=0.1172
p=0.0633
SmallAirway2
LargeAirway2
Trachea
Macrophage2

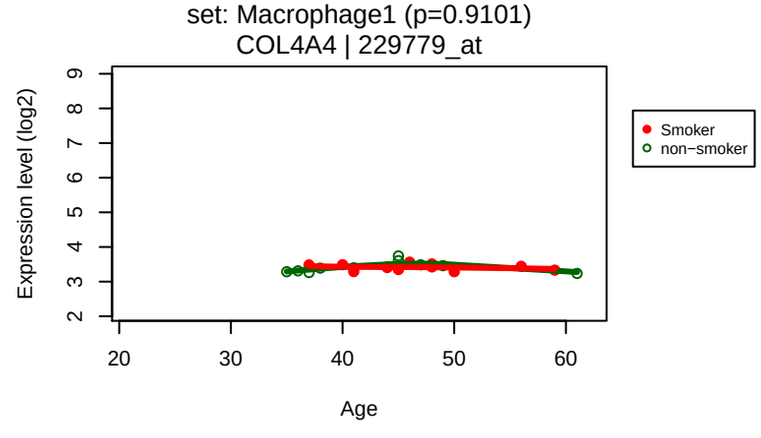

set: SmallAirway1 (p=0.0243)
COL4A4 | 229779_at
Expression level (log2)
Age
Smoker
non−smoker

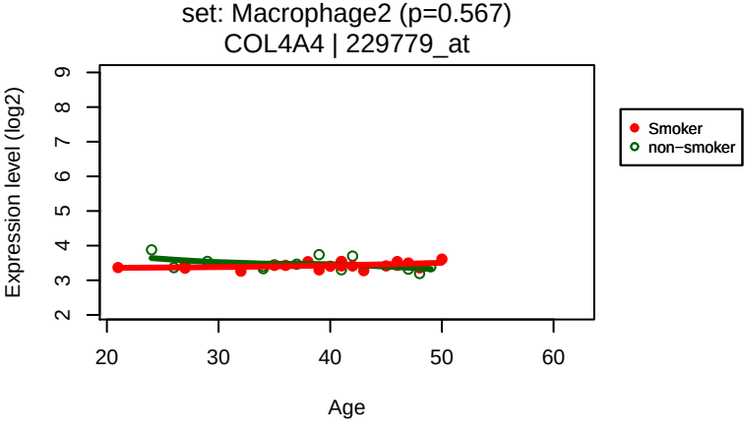

set: SmallAirway2 (p=0.0081)
COL4A4 | 229779_at
Expression level (log2)
Age
Smoker
non−smoker

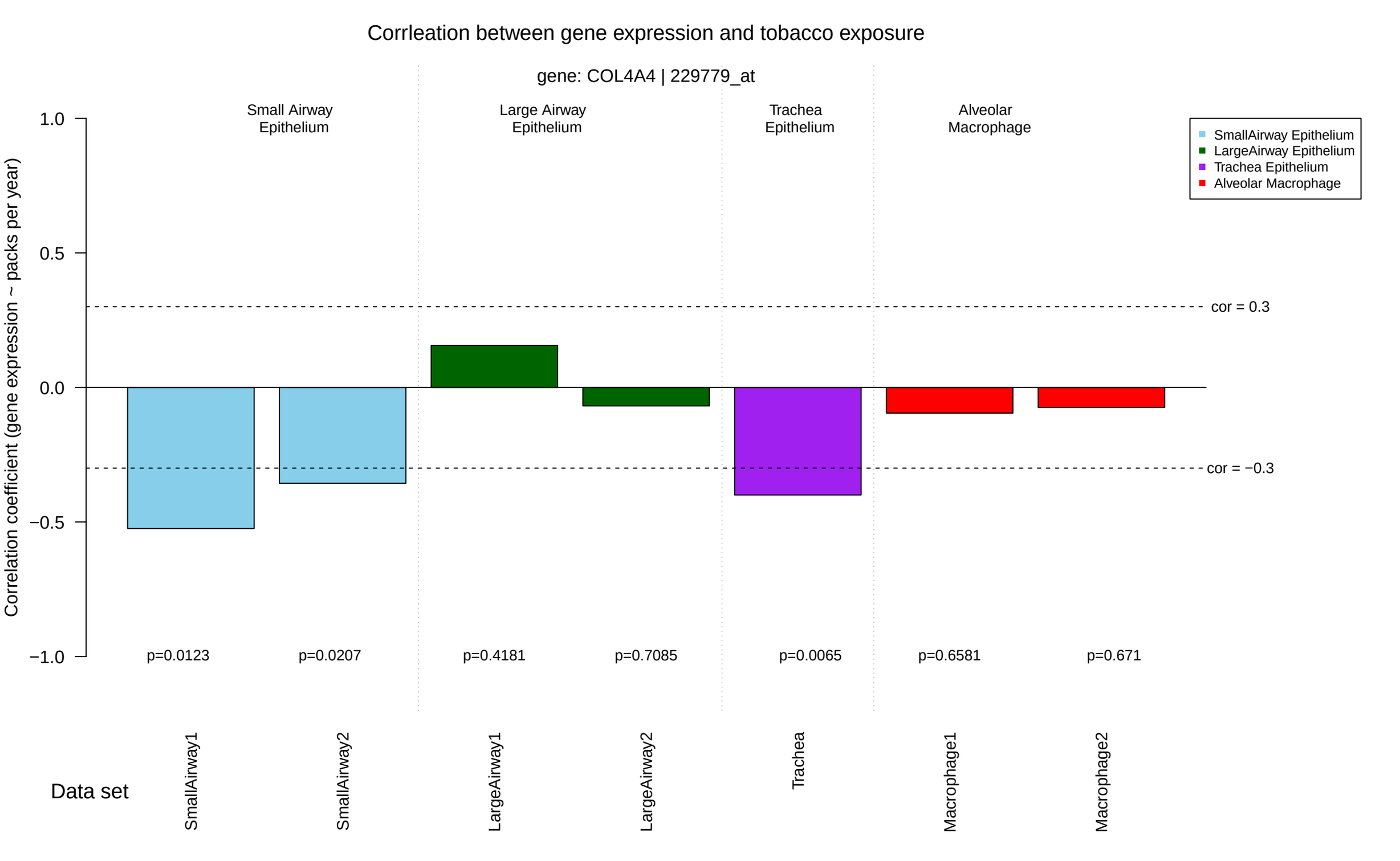

set: LargeAirway1 (p=0.9427)
COL4A4 | 229779_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3899)
COL4A4 | 229779_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0095)
COL4A4 | 229779_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9101)
COL4A4 | 229779_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.567)
COL4A4 | 229779_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A4 | 229779_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0123
p=0.0207
p=0.4181
p=0.7085
p=0.0065
p=0.6581
p=0.671
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2